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The impact of artificial intelligence on economic growth and welfare☆

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ABSTRACT

Focusing on the self-accumulation ability and the nonrival characteristic of artificial intelligence (AI), this paper develops a three-sector endogenous growth model and investigates the impact of the development of AI along the transitional dynamics path and the balanced growth path. The development of AI can increase economic growth along the transitional dynamics path, and can increase household short-run utility if an increase in the accumulation of AI is due to the rising productivity in the goods or AI sector, but can be detrimental to household short-run utility if an increase in the accumulation of AI is because firms use more AI to replace human labor. In addition, the development of AI is not necessarily beneficial to household welfare in the long run. The main results are unaffected when considering the case where AI can improve the accumulation of human capital, the traditional research and development model, and different kinds of physical capital.

1. Introduction

The purpose of this paper is to construct a basic endogenous growth model with artificial intelligence (AI), and to explore the impact of the development of AI on economic growth and welfare. The innovation of this paper is that we incorporate the self-accumulation ability and the nonrival characteristic of AI into the model. In addition, we also consider its ability to replace human labor. The contribution of this paper is that we can investigate the impact of AI accumulation along the transitional dynamics path and in the long run.

AI is considered to have the potential to instigate a fourth industrial revolution,[1] and is dramatically changing people's patterns of interaction and economic activities. The traditional factors of production and physical capital and labor may no longer promote substantial economic growth. It is generally believed that AI will be one of the most important factors determining future economic growth.[2] However, unlike traditional machines, which replaced the use of human and animal labor for simple manual work and heavy or dangerous activities, AI-related inputs may change the type of human work in a comprehensive way.

In contrast to previous industrial revolutions, AI does not simply involve the invention of a new machine or technology. Instead, AI has similarities to the accumulation of human capital, as it can learn and accumulate knowledge by itself. For example, AlphaGo,

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[1] The first industrial revolution, which took place during the 18th and 19th centuries, transformed the mostly agrarian, rural societies into industrial and urban societies. The second industrial revolution, which used electric power to create mass production, took place between 1870 and 1914. The third, also known as the digital revolution, commenced in the 1980s and is ongoing. Its advancements include the personal computer, the Internet, and information and communications technology.

[2] A report developed by Accenture Research in collaboration with Frontier Economics (https://www.accenture.com/us-en/insight-ai-industry-growth) argued that AI could double global gross domestic product (GDP) over the next two decades.

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a computer program that plays the board game Go, which brought widespread attention to AI, can beat human professional Go players. The new successor, AlphaGo Zero, is a version that has been created without using data collected from games played by humans. The programmers of this new version of the game did not use the available accumulated human intelligence used by Alpha Go. Instead, they programmed only the most basic rules of Go into AlphaGo Zero and then allowed it to play games against itself. Everything started from scratch. Only relying on the deep learning and reinforcement of the machine, AlphaGo Zero exceeded the ability of all previous versions of the game in only 40 days.[3]

The above application of AI implies that, by using machines (physical capital, such as computers), human labor (human capital, such as programmers), and AI, the AI can learn by itself. This is the first characteristic of AI that we explore and the innovation in this paper: the self-accumulation ability of AI and its required elements. In addition, AI, as ideas, is a nonrival input. It can be used for production, for automation, and for generating new AI at the same time. AI is similar to a certain technology or knowledge. When this technology is used in the production of final goods, it does not detract from its ability to accumulate AI. This makes AI a different type of capital from physical and human capital, and is the second characteristic of AI that we explore and the innovation in this paper: the nonrival feature of AI.

Nowadays, AI is an important input to production in many industries and is expected to eventually replace human resources. Morikawa (2017) collected original survey data from more than 3000 Japanese firms operating in both the manufacturing and service sectors to investigate their views about the impacts that AI may have on business and employment in the future, and found that firms expect favorable impacts of AI and robotics on their businesses and generally think upgrading human capital is important. That is, the pattern of employment will change in the future. In addition, Acemoglu and Restrepo (2018a) considered the situation in which AI replaces labor, and discussed the impact of automation on labor demand, wages, and employment. They further examined the mismatch between skills and technologies, which will reduce the productivity gains from new technologies. Furthermore, Berg et al. (2018) built models with different elasticities of substitution between robots and labor, and analyzed how automation may transform the labor market in different situations. The above studies suggested that AI may replace human labor, and we thus further consider this feature in this article.[4]

Existing macroeconomic models, however, seldom discuss AI, and therefore, they ignore its self-accumulation ability, which occurs as the machine learns or as people input information into the AI. The endogenous growth models with research and development (R&D) pioneered by Romer (1990) and Aghion and Howitt (1992) usually treat such innovation as an intermediate good produced by a monopolist. However, the characteristics of machines with AI are quite different. They can directly enter the production process and can even replace human labor. In addition, AI, like human capital, can learn and accumulate by itself. Therefore, the traditional R&D model may not fully explain the impact of AI.

Moreover, AI is generally considered important for future economic growth. The growth accounting conducted for the US economy in Fernald and Jones (2014) showed that R&D contribute most to the economy. The authors considered that the possibility of AI allowing machines to replace workers, to some extent, could lead to higher growth in the future. Recent studies, including Acemoglu and Restrepo (2018b), Hémous and Olsen (2016), and Aghion et al. (2017), have discussed the potential implications of AI for the growth process. However, those papers modeled AI in terms of new ideas, as in Romer (1990) and Aghion and Howitt (1992), and they did not take into account the simultaneous contributions to human and physical capital. In addition, Acemoglu and Restrepo (2017) considered low-skill and high-skill automation and discussed wage inequality.

In contrast to the above studies, we introduce AI's ability of self-accumulation and its nonrival characteristic into the (Lucas, 1988) endogenous growth model. We treat AI as an input like human and physical capital, which can directly enter the goods production process and replace workers. People can distribute resources to develop AI, just as in the process of the accumulation of human capital. The innovation contribution of this paper is that we introduce the self-accumulation ability and the nonrival characteristic of AI into an endogenous growth model and explore the impact of the development of AI on macroeconomic performance. In this paper, we investigate the impact of the development of AI using transitional dynamics and the balanced growth path. Exploring the impact along the transitional dynamics path is especially important because it helps us understand and predict the future impact of AI.

The structure of the paper is as follows. In Section 2, we construct a benchmark model and prove the existence of the long-run equilibrium. Section 3 analyzes the impact of AI along the transitional dynamics path and provides some comparative statics analysis in the long run. Section 4 considers the case in which AI can improve the accumulation of human capital. Section 5 compares our benchmark model with the traditional R&D model. In Section 6, we consider different kinds of physical capital in our model. Section 7 offers some concluding remarks.

2. The model

This section builds the basic analytical framework, which draws on the Lucas (1988) endogenous growth model, extended to include the accumulation of AI. We establish a three-sector endogenous growth model. We assume that AI, like human capital, has the ability to accumulate, including through the machine itself learning or through people allocating resources to develop AI. The economy is populated with a continuum of representative households of mass one and a continuum of representative firms of mass one.

[3] For further details about AlphaGo Zero see Silver et al. (2017).

[4] Note that the purpose of this article is to establish a basic model that can explore AI. In order to make the model uncomplicated, the unemployment problem is not discussed in the article, and we leave this for future research. Moreover, Autor (2015) and Pratt (2015) have also discussed the relationship between AI and human jobs.

2.1. Households

The representative household is endowed with one unit of time. At an instant in time, the household devotes the fraction of time u to improving its skills, the fraction of time v to developing AI, and the remaining fraction of time $1-u-v$ to goods production. The household allocates a fraction s of physical capital to develop AI and the remaining fraction $1-s$ to produce goods. We denote k as physical capital, h as human capital, and A as the stock of AI.

We assume that the stock of AI, which is a type of capital, accumulates by itself as a result of its use, just as human capital does. That is, when more resources are devoted to developing AI, higher AI is generated. The process of AI accumulation is as follows:

$$\dot{A} = D(vh)^{\phi}(sk)^{\theta}A^{1-\phi-\theta}, \tag{1}$$

where $\phi, \theta, 1-\phi-\theta \in (0,1)$ and $D > 0$ measures the efficiency of the process of AI accumulation, i.e., the self-accumulation ability of AI.

The development of AI requires machines such as computers, and humans to write programs. Therefore, the accumulation of AI depends on the quantity of resources and time devoted to developing AI, and thus, the accumulation equation of AI contains the physical and human capital allocated to the development of AI. In addition, just like the example, AlphaGo Zero, that we mentioned in the Introduction, AI can accumulate itself by relying on the deep learning and reinforcement of the machine. That is, the above accumulation equation should also include the AI stock itself. Today, the development of AI still requires machines and programmers, but perhaps in the future, AI will develop without heavily relying on humans or old machines, and can accumulate by itself, i.e., $1-\phi-\theta$ may increase and be close to one in the future.

A household's human capital is accumulated via a learning activity, as follows:

$$\dot{h} = Buh, \tag{2}$$

where $B > 0$ measures the efficiency of the process of human capital accumulation. (2) is based on the setting in Lucas (1988).

We use w, r, and r_A to denote the wage rate, the rental rate, and the return to AI, respectively. At any point in time, the representative household's flow budget constraint is:

$$\dot{k} = w(1-u-v)h + r(1-s)k + r_A A + \pi - c, \tag{3}$$

where π is the firm's profits, given that households own the shares of the firm. The budget constraint indicates that unspent income is used to accumulate physical capital. To simplify the model, we assume that the depreciation rates of physical capital, human capital, and AI are zero. Note that the household does not allocate different fractions of AI to different sectors, i.e., AI is a nonrival input. This is because AI is similar to a certain technology or knowledge. When this technology is used in the production of goods, it does not detract from its ability to accumulate AI. This feature also makes AI a different type of capital from physical and human capital.[5]

The household's lifetime utility is represented as:

$$U = \int_{t=0}^{\infty} u(c)e^{-\rho t}dt, \tag{4}$$

where $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, c is consumption, and $\rho > 0$ is the time preference rate. In (4), we use a conventional constant relative risk aversion (CRRA) utility function with a constant intertemporal elasticity of substitution, $1/\sigma$, for consumption.

The representative household's problem is to maximize lifetime utility (4) by choosing between consumption and investment in the goods, education, and AI sectors, subject to the constraints (1), (2), and (3), taking as given the factor prices, r_A, w, and r, and the initial levels of AI, human and physical capital, $A(0)$, $h(0)$, and $k(0)$. Let λ_A, λ_h, and λ denote the Lagrange multipliers associated with (1), (2), and (3), respectively. The first-order conditions are as follows:

$$c^{-\sigma} = \lambda, \tag{5a}$$

$$\lambda_h B = \lambda w, \tag{5b}$$

$$\lambda_A \phi D(vh)^{\phi-1}(sk)^{\theta}A^{1-\phi-\theta} = \lambda w, \tag{5c}$$

$$\lambda_A \theta D(vh)^{\phi}(sk)^{\theta-1}A^{1-\phi-\theta} = \lambda r, \tag{5d}$$

$$\dot{\lambda}_h - \rho\lambda_h = -[\lambda_h Bu + \lambda_A \phi D v^{\phi} h^{\phi-1}(sk)^{\theta}A^{1-\phi-\theta} + \lambda w(1-u-v)], \tag{5e}$$

$$\dot{\lambda}_A - \rho\lambda_A = -[\lambda_A(1-\phi-\theta)D(vh)^{\phi}(sk)^{\theta}A^{-\phi-\theta} + \lambda r_A], \tag{5f}$$

$$\dot{\lambda} - \rho\lambda = -[\lambda_A \theta D(vh)^{\phi}s^{\theta}k^{\theta-1}A^{1-\phi-\theta} + \lambda r(1-s)]. \tag{5g}$$

We also have the following transversality conditions: $\lim_{t\to\infty} e^{-\rho t}\lambda(t)k(t) = 0$, $\lim_{t\to\infty} e^{-\rho t}\lambda_h(t)h(t) = 0$, and $\lim_{t\to\infty} e^{-\rho t}\lambda_A(t)A(t) = 0$.

[5] Our model can further study the case that the household allocates different fractions of AI to different sectors. This makes the model complicated; however, our main conclusion is unchanged. In addition, if the process of human capital accumulation also depends on physical capital, i.e., a fraction of physical capital is used to develop human capital, which is the setting in Bond et al. (1996) and Mino (2001), our results still hold.

By using (5b)–(5d), we can rewrite (5e)–(5g) as the following equations:

$$\frac{\dot{\lambda}_h}{\lambda_h} = \rho - B, \tag{6a}$$

$$\frac{\dot{\lambda}}{\lambda} = \rho - r, \tag{6b}$$

$$\frac{\dot{\lambda}_A}{\lambda_A} = \rho - D(vh)^{\phi}(sk)^{\theta}A^{-\phi-\theta}(1-\phi-\theta+\frac{r_A\phi A}{wvh}). \tag{6c}$$

Moreover, combining (5c) and (5d) yields the following relationship:

$$\frac{\phi}{\theta}\frac{sk}{vh} = \frac{w}{r}. \tag{6d}$$

2.2. Firms

The representative firm produces a single final good y by renting physical capital and AI and employing labor under the following production technology:

$$y = F[(1-s)k]^{1-\alpha}\{(1-a)[(1-u-v)h]^{\beta} + aA^{\beta}\}^{\frac{\alpha}{\beta}}, \tag{7}$$

where $F > 0$ is productivity, $\alpha \in (0,1)$ is the labor share, and thus $1-\alpha \in (0,1)$ is the physical capital share. As presented in the Introduction, AI is expected to eventually replace human resources. Therefore, we use a constant elasticity of substitution (CES) production function where AI and human labor can replace each other. Note that $\beta = 1 - 1/\beta_{h,A} \in (-\infty, 1]$, in which $\beta_{h,A} \geq 0$ is the elasticity of substitution between human labor and AI, and $a \in (0,1)$ is the share parameter. The bigger a is, the more AI is used (compared with human labor) for production.

Note that when $a = 0$, our model is the Lucas (1988) two-sector endogenous growth model. This is because the households have no incentive to develop AI because it does not help with the production of goods in this situation.

The firm's objective is to choose inputs to maximize the following profits:

$$\pi = y - w(1-u-v)h - r(1-s)k - r_A A. \tag{8}$$

The first-order conditions are as follows:

$$(1-\alpha)\frac{y}{(1-s)k} = r, \tag{9a}$$

$$\alpha\frac{y}{(1-u-v)h}\frac{(1-a)[(1-u-v)h]^{\beta}}{(1-a)[(1-u-v)h]^{\beta} + aA^{\beta}} = w, \tag{9b}$$

$$\alpha\frac{y}{A}\frac{aA^{\beta}}{(1-a)[(1-u-v)h]^{\beta} + aA^{\beta}} = r_A. \tag{9c}$$

2.3. Equilibrium

An equilibrium consists of the time paths of the households' choices, the firms' choices, and prices, such that: (1) households optimize; (2) firms optimize; and (3) all markets clear.

In equilibrium, by combining (3) and (8), along with (9a)–(9c), we can derive the aggregate goods market clearance constraint as follows:

$$\frac{\dot{k}}{k} = F(1-s)^{1-\alpha}(\frac{k}{h})^{-\alpha}[(1-a)(1-u-v)^{\beta} + a(\frac{A}{h})^{\beta}]^{\frac{\alpha}{\beta}} - \frac{c/h}{k/h}. \tag{10}$$

Moreover, combining (5a) and (6b), along with (9a), yields:

$$\frac{\dot{c}}{c} = \frac{1-\alpha}{\sigma}F(1-s)^{-\alpha}(\frac{k}{h})^{-\alpha}[(1-a)(1-u-v)^{\beta} + a(\frac{A}{h})^{\beta}]^{\frac{\alpha}{\beta}} - \frac{\rho}{\sigma}. \tag{11}$$

Furthermore, using (6d), (9a), and (9b), we can obtain the following relationship:

$$\frac{\phi}{\theta}\frac{s}{v} = \frac{\alpha(1-s)}{1-\alpha}[(1-a)(1-u-v)^{\beta} + a(\frac{A}{h})^{\beta}]^{-1}(1-a)(1-u-v)^{\beta-1}. \tag{12}$$

In addition, using (1) and (2) we derive that: $\frac{\dot{h}}{h} = Bu$ and $\frac{\dot{A}}{A} = Dv^{\phi}s^{\theta}(\frac{k}{h})^{\theta}(\frac{A}{h})^{-\phi-\theta}$, respectively. By differentiating (5b)–(5d) with respect to time, along with (6a)–(6c), we can derive the dynamic equations of $\frac{\dot{u}}{u}$, $\frac{\dot{v}}{v}$, and $\frac{\dot{s}}{s}$, which are functions of u, v, s, $\frac{A}{h}$, $\frac{k}{h}$, and $\frac{c}{h}$. To analyze the equilibrium, we transform the perpetually growing variables of consumption, physical capital, AI, and human capital into the ratios c/h, k/h, and A/h. Denote $x \equiv \frac{A}{h}$, $q \equiv \frac{k}{h}$, and $z \equiv \frac{c}{h}$. By using the dynamic equations of $\frac{\dot{u}}{u}$, $\frac{\dot{v}}{v}$, $\frac{\dot{s}}{s}$, (1), (2), (10), and (11), we can derive the time paths of u, v, s, x, q, and z.[6]

6 $\frac{\dot{x}}{x} = \frac{\dot{A}}{A} - \frac{\dot{h}}{h}$, $\frac{\dot{q}}{q} = \frac{\dot{k}}{k} - \frac{\dot{h}}{h}$, and $\frac{\dot{z}}{z} = \frac{\dot{c}}{c} - \frac{\dot{h}}{h}$. Other calculation details are available on request.

2.4. The balanced growth path

Now, we analyze the existence of the balanced growth path (BGP). A long-run equilibrium is a BGP along which the household's fractions of time devoted to learning and developing AI, u and v, respectively, and the household's fraction of physical capital s allocated to developing AI are all constant, and consumption c, physical capital k, human capital h, and AI A all grow at the same rate, denoted by γ. Note that (12) shows that A and h have the same growth rate in the long run. That is, (11) implies that k and h have the same growth rate in the long run, as do k and y, and (10) shows that this is also the case for c and h.

In the long run, by using $\frac{\dot{z}}{z} = \frac{\dot{q}}{q} = \frac{\dot{x}}{x} = \frac{\dot{u}}{u} = \frac{\dot{v}}{v} = \frac{\dot{s}}{s} = 0$, we can derive the following relationships along the BGP:

$$u = \frac{B-\rho}{\sigma B}, \tag{13a}$$

$$x = [(\frac{1}{u} - 1 + \phi + \theta)\frac{1-a}{a}(1-u-v)^{\beta-1}\frac{v}{\phi}]^{1/\beta}, \tag{13b}$$

$$s = \frac{\frac{\alpha}{1-\alpha}}{\frac{\phi}{\theta}[\frac{1-u-v}{v} + (\frac{1}{u} - 1 + \phi + \theta)\frac{1}{\phi}] + \frac{\alpha}{1-\alpha}} \in (0,1), \tag{13c}$$

$$q = (\frac{B}{D}uv^{-\phi}s^{-\theta}x^{\phi+\theta})^{\frac{1}{\theta}}, \tag{13d}$$

$$z = (\frac{1-s}{1-\alpha}B - Bu)q, \tag{13e}$$

$$B = (1-\alpha)F[(1-s)q]^{-\alpha}[\frac{\theta v\alpha(1-s)(1-a)}{\phi s(1-\alpha)(1-u-v)^{1-\beta}}]^{\frac{\alpha}{\beta}}. \tag{13f}$$

Under the conditions that $\sigma B > B - \rho > 0$, (13a) implies that there exists a unique level of $u \in (0,1)$ in the long run. That is, (13b)–(13e) imply that x, s, q, and z are functions of v. Moreover, (13c) implies that s is increasing in v from 0 when $v = 0$ to a constant between zero and one when $v = 1-u$. We can derive the long-run level of v by using (13f).

Furthermore, after some sorting, along with (13c) and (13d), (13f) becomes:

$$B = (1-\alpha)F(\frac{Bu}{D})^{-\frac{\alpha}{\theta}}[\frac{\alpha(1-a)\theta}{(1-\alpha)\phi}]^{\frac{\alpha}{\beta}}[\frac{1-a}{a\phi}(\frac{1}{u} - 1 + \phi + \theta)]^{\frac{-\alpha}{\beta}(1+\frac{\phi}{\theta})}(\frac{s}{1-s})^{\alpha\frac{\beta-1}{\beta}}(\frac{1-u-v}{v})^{\frac{1-\beta}{\beta}\frac{\phi\alpha}{\theta}}. \tag{13g}$$

The left-hand side of (13g) is a constant. Remember that $\beta \le 1$. When $\beta > 0$, the right-hand side of (13g) is decreasing in v from infinity when $v = 0$ to zero when $v = 1-u$. Therefore, the constant left-hand side of (13g) and the negatively sloping right-hand side of (13g) must intersect. When $\beta < 0$, the right-hand side of (13g) is increasing in v from zero when $v = 0$ to infinity when $v = 1-u$. Therefore, the constant left-hand side of (13g) and the positively sloping right-hand side of (13g) must intersect. Thus, there exists a unique BGP along which $v \in (0, 1-u)$. Then, we can derive the unique long-run levels of x, s, q, and z from (13b)–(13e).

The growth rate of the economy, which is $\gamma = \frac{\dot{y}}{y} = \frac{\dot{h}}{h} = \frac{\dot{k}}{k} = \frac{\dot{A}}{A} = \frac{\dot{c}}{c}$, can be derived as follows:

$$\gamma = Bu. \tag{13h}$$

Household welfare in the long run can be derived as follows:

$$U = \frac{z^{1-\sigma}h(0)^{1-\sigma}}{1-\sigma}\frac{1}{\rho-\gamma(1-\sigma)} - \frac{1}{\rho(1-\sigma)}. \tag{13i}$$

3. Numerical analysis

To more clearly discuss the effect of AI along the transitional dynamics and balance growth paths, we undertake a numerical analysis. Below, we will first introduce how we calibrate our model to fit the actual data, then we will investigate the impact of key variables along the transitional dynamic paths, and then we will study the comparative statics of each relevant variable in the long run.

3.1. Calibration

To quantify the results, we calibrate this three-sector endogenous growth model along the BGP to reproduce the key features of the US economy at annual frequencies. We use data for the period 2000–2017. According to Penn World Tables (version 9.1), the GDP growth rate, the consumption–output ratio, and human capital index in the US during 2000–2017 were 2.0234%, 0.6748, and 3.6688, respectively.[7] As that human capital index is based on years of schooling and returns to education, we assume the index is the human capital to output ratio, h/y. Thus, we set initial $\gamma = 2.0234\%$, $c/y = 0.6748$, and $h/y = 3.6688$. By using the same database, we can calculate $q = 0.8906$ and $z = 0.1839$.

In addition, it is more common to use a capital share of output between 0.30 and 0.40 (see the examples in Cooley (1995)); therefore, we set the capital share in the production function of final goods at 0.3, and thus $\alpha = 1 - 0.3 = 0.7$. Following Lucas (1990), we set $\sigma = 2$. Moreover, Kydland and Prescott (1991) used 4% as the annual rate of time preference, so we set $\rho = 4\%$. By

[7] Data are from https://www.rug.nl/ggdc/productivity/pwt/.

Table 1
Calibration results of the benchmark model.

u	v	s	x	q	z	γ	U
(A) *The benchmark case*							
0.2515	0.2644	0.1546	0.1428	0.8906	0.1839	0.0202	−65.2577
(B) F *is increasing from* 0.6640 *to* 0.7304							
0.2515	0.2525	0.1532	0.1535	1.0877	0.2250	0.0202	−48.7734
(C) D *is increasing from* 0.0167 *to* 0.0183							
0.2515	0.2398	0.1517	0.1659	1.0157	0.2106	0.0202	−53.8405
(D) a *is increasing from* 0.5 *to* 0.55							
0.2515	0.2828	0.1565	0.1466	0.8671	0.1786	0.0202	−67.9312

using (13a) and (13h), we can calibrate $u = 0.2515$ and $B = 0.0805$. Note that the growth rate of the economy is $\gamma = \frac{\dot{y}}{y} = \frac{\dot{h}}{h} = \frac{\dot{k}}{k} = \frac{\dot{c}}{c}$ in the long run.

To date, there are no data for the elasticity of substitution between human labor and AI. According to Chirinko and Mallick (2017), the preferred point estimate of the elasticity of substitution between labor and capital is around 0.406. We can calibrate $\beta = -1.4631$.[8] Assume that the capital, labor, and AI shares in the production function of AI accumulation are equal, i.e., $\phi = \theta = 1/3$, and the CES share parameters of AI and human labor in the production function of final goods are equal, i.e., $a = 0.5$. We can calibrate $s = 0.1546$, $v = 0.2644$, $x = 0.1428$, $D = 0.0167$, and $F = 0.6640$ by using (13b)–(13f). Finally, by normalizing the initial human capital level at $h(0) = 1$, the household's welfare can be derived from (13i) as $U = -65.2577$. We summarize the related macroeconomic variables in Table 1 (case A).

Although the parameters related to AI are given directly, using different parameters provides consistent results. We conduct long-term comparative static analysis below, which is a type of sensitivity analysis for this. In addition, the above simulation only analyzed long-term results; today, however, programmers around the world are working hard to develop AI. Therefore, studying the transitional dynamics of the development of AI is also very important to help us understand the current environment, which is addressed in the next section.

3.2. Transitional dynamics

In this section, we examine the impact of changing the parameters related to AI along the transitional dynamics path. In this paper, we focus on the self-accumulation ability and the nonrival characteristic of AI, and the fact that AI can replace human labor in the production of goods. That is, we investigate the transition impact of an increase in the parameter related to productivity and the CES share of AI in the production of goods, and the self-accumulation ability of AI.

From the above calculations, our model can be organized into six equilibrium equations, $\frac{\dot{z}}{z}$, $\frac{\dot{q}}{q}$, $\frac{\dot{x}}{x}$, $\frac{\dot{u}}{u}$, $\frac{\dot{v}}{v}$, and $\frac{\dot{s}}{s}$. By using (12), we can simplify the above equilibrium conditions into a five-dimensional dynamical system with state vector (z, q, x, u, v). To investigate the dynamic effects of the invention of AI, we start by taking a linear Taylor's expansion of the dynamical system in the neighborhood of the unique BGP and obtain a Jacobean matrix.

We must find two negative and three positive eigenvalues associated with the Jacobean matrix in order to guarantee a unique equilibrium path toward the BGP. The values of $z(t), q(t), x(t), u(t)$, and $v(t)$ in the unique equilibrium path are then each represented by the sum of their own new BGP, which is solved from the three-sector endogenous growth model, and two products of three components: (i) a coefficient, (ii) an exponential to the power of the negative eigenvalue multiplied by time t, and (iii) the corresponding eigenvector of the negative eigenvalue. The coefficient is determined by the boundary conditions in which the state variables, q and x, are predetermined. Then, the values of $s(t)$ are derived from (12).[9]

We first discuss the transition results from a 10% increase in productivity in the goods sector in which F ranges from 0.6640 to 0.7304, i.e., the transitional dynamics of case (B) in Table 1 with case (A) as the initial point. The results are shown in Fig. 1. To clearly understand the impact of the moment when F is increased, we only show the changes for the first 10 periods in the graph of the discounted utility and the changes for the first 100 periods in other graphs in Fig. 1.

Fig. 1 implies that when the productivity of goods production increases, the household has incentives to allocate more resources, including working time and physical capital stock, to the goods sector. That is, u, v, and s decrease, i.e., $1 - u - v$ and $1 - s$ increase, at the moment when F increases. Because the production of goods is higher and less time is allocated to accumulate human capital in the short term, the ratio of consumption to human capital (z) increases, and the economic growth rate ($\dot{y}/y$) is also higher in the short run, as is the short-run utility.

As the production of products requires the help of AI, the amount of short-term resources allocated to the AI sector will increase. Fig. 1 shows that the short-term levels of v and s will overshoot their long-term levels. Therefore, the ratio of AI to human capital

[8] The model displays similar qualitative features for alternative values of β.

[9] Details of the linear Taylor's expansion used to calculate the dynamic paths are available on request. The BGP is a saddle in all our numerical exercises.

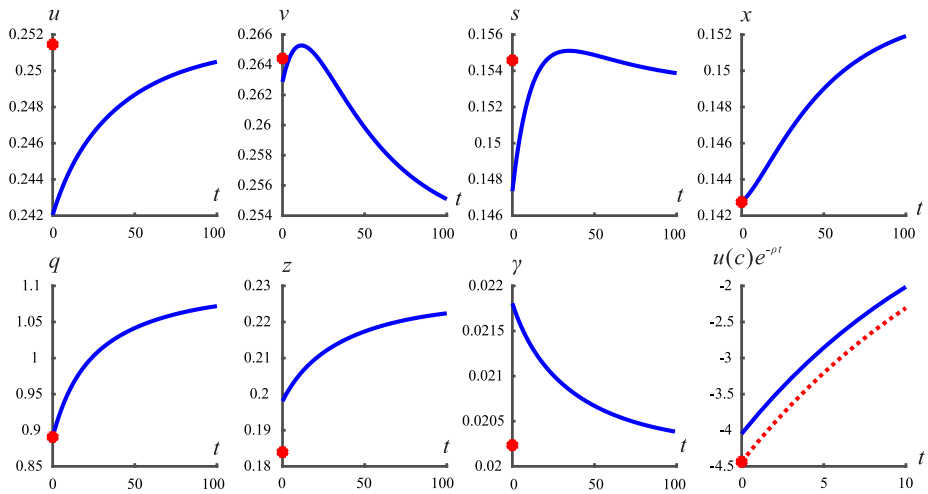

Fig. 1. Transitional dynamics under an increase in productivity in the goods sector. Note: The initial point at time 0 is the benchmark case, i.e., case (A) in Table 1. The dotted line in the rightmost graph in the second row is the discounted utility in the benchmark case.

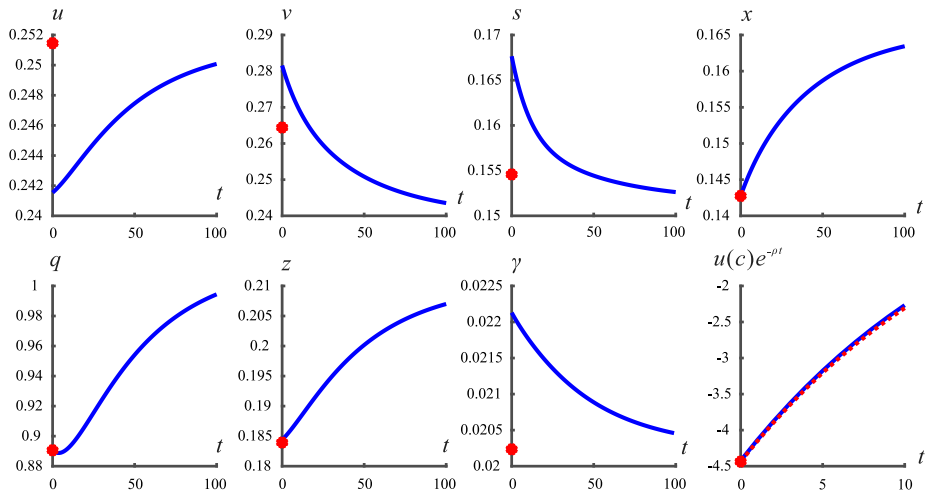

Fig. 2. Transitional dynamics under an increase in AI self-accumulation abilities. Note: Please refer to Fig. 1.

increases over time, as do the ratios of physical capital and consumption to human capital because of a greater production of goods. Briefly, the accumulation of AI contributes to economic growth and public welfare along the transitional dynamics path.

We now investigate the transition impact from a 10% increase in AI self-accumulation abilities in which D ranges from 0.0167 to 0.0183, i.e., the transitional dynamics of case (C) in Table 1 with case (A) as the initial point. The results are shown in Fig. 2. As the efficiency of AI accumulation increases, the household will allocate more time and resources to develop AI, and thus v and s increase immediately. Therefore, the ratio of AI to human capital increases over time. Because AI can help the production of output, short-run economic growth increases even though less human capital is accumulated in the short run because the household allocates less time to learning (u decreases immediately). Therefore, the ratios of AI, physical capital, and consumption to human capital increase over time because of less human capital accumulation.

However, as more resources are allocated to developing AI and less time is allocated to work, in order to not reduce consumption (Fig. 2 shows that z increases slightly at once), the household reduces the accumulation of physical capital in the short term, and thus q decreases slightly in the short term but keeps rising over time. As the short-term z and the economic growth rate are higher than those in the benchmark case, the short-term utility is also higher than its benchmark level. In addition, similar to Fig. 1, Fig. 2 shows that the short-term levels of v and s overshoot their long-term levels. Under a higher AI self-accumulation ability, the accumulation of AI contributes to economic growth and public welfare along the transitional dynamics path.

We now investigate the transition impact from a 10% increase in the CES share of AI in the final goods production in which a ranges from 0.5 to 0.55, i.e., the transitional dynamics of case (D) in Table 1 with case (A) as the initial point. The results are shown in Fig. 3. As the production of final goods needs more AI, the household allocates more time and resources to develop AI,

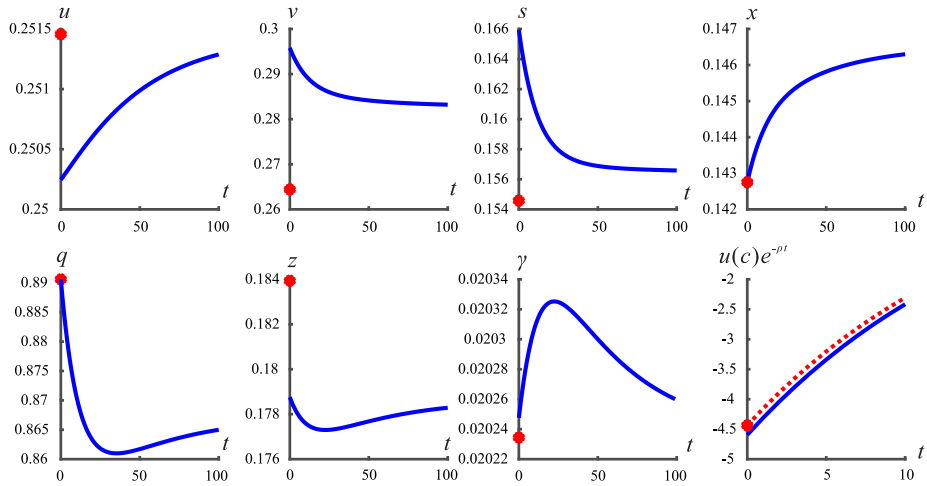

Fig. 3. Transitional dynamics under an increase in the CES share of AI in goods production. Note: Please refer to Fig. 1.

Table 2
Comparative statics analysis in the long run.

	u	v	s	x	q	z	γ	U
B	+	+	+	−	−	−	+	−
F	0	−	−	+	+	+	0	+
D	0	−	−	+	+	+	0	+
a	0	+	+	+	−	−	0	−
ϕ	0	+	−	+	+	+	0	+
θ	0	−	+	+	+	+	0	+
β	0	−	−	−	−	−	0	−

Note: + and − indicate that the effects of changing the parameters on the related variables are monotonically increasing and decreasing, respectively. 0 represents that the change in the parameter value does not affect the value of the related variables.

and thus v and s increase immediately. Therefore, the time that the household allocates to learning and working decreases. That is, the ratio of AI to human capital increases over time.

With the help of AI, the goods production increases even though the working time decreases, so the short-run economic growth rate increases. Fig. 3 shows that the short-term levels of v and s and the economic growth rate overshoot their long-term levels. However, as more resources are allocated to accumulate AI, the ratios of physical capital and consumption to human capital decrease, as does the level of short-term utility. That is, even if the development of AI is helpful for short-term economic growth, people may not feel happier along the transitional dynamics path.

In summary, along the transitional dynamics path, the development of AI does help increase the short-term economic growth rate, but it does not necessarily improve the welfare of the people. If an increase in the development of AI is from the improvement of production efficiency, whether it is from the goods or the AI sector, the accumulation of AI is good for the welfare of the people. However, if an increase in the development of AI is from a change in production methods and firms use AI to replace human labor, the accumulation of AI will reduce the welfare of the people.

Note that using different parameter values results in a consistent outcome. Here, we only focus on the characteristics of AI. For brevity, we only illustrate three cases in Figs. 1–3, and list the results of changing the parameter values in the next section by using long-term comparative static analysis. Note that whenever a parameter is changed, there will be different time paths.

3.3. Comparative static analysis in the long run

We now investigate the long-run effects of different parameter values on those variables related to AI, economic growth, and welfare. The results are a type of robustness check of effects of the development of AI in our model. The values of all variables along the BGP are derived from (13a)–(13i). We discuss the long-run impact of changing productivity in three sectors (B, F, and D), the CES share of AI in the final goods production (a), the physical capital and labor shares in the AI sector (ϕ and θ), and the elasticity of substitution between human labor and AI (β). In the following comparative static exercises, we use the parameter settings in case (A) in Table 1 as the benchmark, and then change only one parameter at a time. We list all comparative statics results in Table 2.

Note that the household's time allocated to the accumulation of human capital (u) and the economic growth rate along the BGP are unchanged under all cases except for changing B. This is because we use the standard human capital accumulation function as in Lucas (1988), and the growth rates of y, k, h, A, and c are the same and equal to (13h) in the long run. An increase in efficiency in the education sector increases the household's incentive to devote more time to learning and thus, the long-run value

of u increases, as does the growth rate of the economy. Because the accumulation of human capital contributes to the development of AI, the household has more incentive to allocate more time and resources to develop AI and thus, v and s increase. However, greater human capital accumulation will reduce the ratios of AI, physical capital, and consumption to human capital, and thus, household welfare falls as B increases. This implies factors that are helpful to economic growth are not necessarily good for the welfare of the people.

Regarding an increase in productivity levels in the goods sector (F), the household allocates more resources to the production of goods, leading to a reduction in the amount of resources in the AI sector and thus, v and s decrease. Physical capital accumulation increases because of higher productivity. Thus, the ratio of physical capital to human capital increases, and this increases the accumulation of AI. Because of the higher production of goods, the ratio of consumption to human capital increases, as does household welfare.

In addition, an increase in the self-accumulation ability of AI (D) has two effects on household resource allocation. One is that it enhances the household's incentives to increase the resource allocation in the AI sector, and the other is that the household can develop more AI without spending too much time and physical capital in the AI sector because of the high cumulative efficiency of AI. Table 2 shows that the latter effect dominates, and thus v and s decrease as D increases. However, the ratio of AI to human capital is still increasing in D because of higher cumulative efficiency, which can help the production of goods, as do the ratios of physical capital and consumption to human capital and household welfare.

With regard to an increase in the CES share of AI in the goods production function, the household will devote more time and resources to developing AI when AI becomes more important for production. That is, the ratio of AI to human capital is increasing in a. However, although the cumulative performance of AI helps production, too few resources are allocated to the production of goods, which reduces the ratios of physical capital and consumption to human capital, as does household welfare. Therefore, the development of AI is not necessarily good for the welfare of the people.

Moreover, the households devote more time to developing AI, yet reduce the fraction of physical capital allocated to the AI sector when the labor share in the AI accumulation function increases. In this case, in the AI sector, the positive effect of more labor (human capital) dominates the negative effect of less physical capital and thus, the ratio of AI to human capital increases as ϕ increases. Furthermore, the household allocates a higher fraction of physical capital to the AI sector, yet it devotes less time to developing AI when the capital share in the AI accumulation function increases. Similarly, the positive effect of more physical capital dominates the negative effect of less labor (human capital) and thus, the ratio of AI to human capital increases as θ increases. The effects of an increase in ϕ or θ on q, z, and household welfare are similar to that under an increase in D.

Furthermore, Table 2 shows that household welfare decreases as the elasticity of substitution between human labor and AI increases. The comparative statics results of changing β and a imply that when the degree of mutual substitution between AI and human labor is higher or firms use more AI to replace human labor, it may be detrimental to the welfare of the people. That is, the development of AI is not necessarily beneficial for household welfare.

Note that we use a conventional CRRA utility function in this paper and thus, there is a direct positive correlation between household welfare and the ratio of consumption to human capital. Moreover, even if AI replaces human labor in the production of goods, the households will still accumulate human capital because it is an indispensable element in the development of AI. The above results also imply that an individual with knowledge or skills related to AI is the type of worker who will be needed in the future.

Finally, we perform further robustness checks. We change one of the parameters a, F, D, B, ϕ, θ, and β slightly at one time, and redo the above comparative statics exercises. For example, if we increase the level of a from 0.5 to 0.55, the comparative statics results are the same as those in Table 2. The same results are obtained when we change the level of F, D, B, ϕ, θ, and β. Therefore, our numerical exercises are robust.

3.4. Split the sample period into a pre-AI-rise period and a post-AI-rise period

AI intensity has been changing over the past couple of decades. It is thus valuable to divide the quantitative study into two regimes: low and high AI-intensity regimes, based on AI adoption. As the term Industry 4.0 (the fourth industrial revolution) originated in 2011 from a project in the high-tech strategy of the German government, we split the sample period (2000–2017) into 2000–2010 and 2011–2017 which are the pre-AI-rise period and the post-AI-rise period, respectively.

By using the same database as in Section 3.1, the GDP growth rate, the consumption–output ratio, and human capital index in the US during 2000–2010 were 1.9693%, 0.6724, and 3.6344, respectively, and those during 2011–2017 were 2.1085%, 0.6786, and 3.7228, respectively. By using the same settings and calibration steps as those in Section 3.1, we can calibrate $B = 0.0794$, $D = 0.0184$, and $F = 0.6081$ during 2000–2010, and $B = 0.0822$, $D = 0.0144$, and $F = 0.7617$ during 2011–2017. Related macroeconomic variables are summarized in Table 3 (case A for the period 2000–2010 and case E for the period 2011–2017).

By comparing the resource and time allocation between the two periods, we obtain that the household does devote more time and physical capital to accumulating human capital and developing AI during the post-AI-rise period, i.e., the levels of u, v, and s during the period 2011–2017 are higher than those during the period 2000–2010. Economic growth and household welfare are also higher during the period 2011–2017. We also list the long-run simulation results under a 10% increase in productivity in the goods sector (F), in AI self-accumulation abilities (D), and in the CES share of AI in goods production (a) in Table 3 (cases B–D for the period 2000–2010 and cases F–H for the period 2011–2017). As for the transitional dynamics and the long-run effects of different parameter values, both periods obtain the same results as those in Figs. 1–3 and Table 2. That is, splitting the sample period does not change our results, while we find that the investment in the development of AI is increasing over time.

Table 3
Calibration results during pre- and post-AI-rise periods.

u	v	s	x	q	z	γ	U
Cases during the period 2000–2010							
(A) *The benchmark case*							
0.2481	0.2422	0.1502	0.1637	0.9018	0.1850	0.0197	−65.5443
(B) F *is increasing from* 0.6081 *to* 0.6690							
0.2481	0.2309	0.1488	0.1754	1.0970	0.2255	0.0197	−49.2906
(C) D *is increasing from* 0.0184 *to* 0.0202							
0.2481	0.2190	0.1471	0.1890	1.0201	0.2101	0.0197	−54.7218
(D) a *is increasing from* 0.5 *to* 0.55							
0.2481	0.2596	0.1523	0.1688	0.8832	0.1807	0.0197	−67.6891
Cases during the period 2011–2017							
(E) *The benchmark case*							
0.2566	0.2985	0.1608	0.1157	0.8731	0.1823	0.0211	−64.8100
(F) F *is increasing from* 0.7617 *to* 0.8379							
0.2566	0.2858	0.1595	0.1250	1.0726	0.2243	0.0211	−47.9882
(G) D *is increasing from* 0.0144 *to* 0.0159							
0.2566	0.2721	0.1581	0.1358	1.0080	0.2112	0.0211	−52.5172
(H) a *is increasing from* 0.5 *to* 0.55							
0.2566	0.3179	0.1625	0.1179	0.8426	0.1755	0.0211	−68.2668

Table 4
Calibration results for the model where AI can improve the accumulation of human capital.

u	v	s	x	q	z	γ	U
(A) *The benchmark case*							
0.1257	0.2644	0.1546	0.2171	0.8906	0.1839	0.0202	−65.2577
(B) F *is increasing from* 0.5058 *to* 0.5564							
0.1271	0.2592	0.1555	0.2246	1.0345	0.2156	0.0207	−51.4111
(C) D *is increasing from* 0.0221 *to* 0.0243							
0.1287	0.2535	0.1564	0.2330	0.9160	0.1929	0.0212	−59.7063
(D) a *is increasing from* 0.5 *to* 0.55							
0.1264	0.2873	0.1576	0.2205	0.8564	0.1770	0.0204	−68.4509

4. AI improves the accumulation of human capital

In the benchmark model, we focus on the process of AI accumulation and its ability to replace human labor. Here, we consider the case where AI can improve the accumulation of human capital, and investigate how this channel affects the growth and welfare implications.

Similar to the setting of (1), the human capital accumulation function becomes as follows:

$$\dot{h} = B(uh)^{\eta} A^{1-\eta}, \tag{14}$$

where $\eta \in (0, 1)$. Other model settings are the same as those in the benchmark model. That is, by using the same steps as in the benchmark model, we can derive the equilibrium conditions and the BGP. Calculation details are in the online appendix. To more clearly discuss the effect of AI along the transitional dynamics and balanced growth paths, and to compare it with the benchmark model, we perform a numerical analysis.

To quantify the results, we calibrate the model along the BGP to reproduce the key features of the US economy at annual frequencies. Similarly, we use data for the period 2000–2017. By using the same database and settings as those in the benchmark model (Section 3.1), and assuming that $\eta = 0.5$, we can calibrate $B = 0.1225$, $D = 0.0221$, and $F = 0.5058$. Related macroeconomic variables are summarized in Table 4 (case A).

We first discuss the impact of changing the parameters related to AI along the transitional dynamics path. To compare this impact with the benchmark model, here we investigate the transition impact of an increase in the parameter related to productivity and the CES share of AI in the production of goods, as well as the self-accumulation ability of AI. We increase one parameter by 10% at a time. The long-run simulation results are listed in Table 4 (cases B–D, respectively).

Table 5
Calibration results of the model with conventional R&D.

u	v	d	s	x	q	z	p	γ	U
(A) *The benchmark case*									
0.2515	0.2004	0.0578	0.1264	0.1763	0.8906	0.1839	0.00051	0.0202	−65.2577
(B) G *is increasing from* 0.0019 *to* 0.0021									
0.2515	0.1989	0.0575	0.1262	0.1783	0.9202	0.1901	0.00062	0.0202	−62.3280
(C) D *is increasing from* 0.0225 *to* 0.0248									
0.2515	0.1833	0.0542	0.1236	0.2009	0.9718	0.2014	0.00048	0.0202	−57.4307
(D) a *is increasing from* 0.5 *to* 0.55									
0.2515	0.2138	0.0606	0.1283	0.1828	0.8851	0.1824	0.00054	0.0202	−66.0337

The transitional dynamics paths for variables under 10% increases in F, D, and a are similar to those in Figs. 1–3, respectively.[10] That is, the case where AI can improve the accumulation of human capital does not change our main results: the development of AI can increase economic growth along the transitional dynamics path, and can benefit the household's short-run utility if the accumulation of AI is due to the increasing productivity in the goods or AI sector, but can be detrimental to the household's short-run utility if the accumulation of AI is because firms use more AI to replace human labor.

As for the long-run effects of different parameter values on those variables related to AI, economic growth, and welfare. Differing from the results in Table 2, the household's time allocated to the education sector (u) is changed under different parameters as AI can now improve the accumulation of human capital.[11] However, the main conclusion still holds in this case, where the impact of such parameters on the economic growth rate mainly depends on that impact on u. Overall, the situation whereby AI can improve the accumulation of human capital does not change our main results.

5. Comparison with the conventional R&D model

In this paper, we treat AI as an input like physical and human capital that directly enters the production process, and this is very different from the conventional R&D model that usually treats such innovation as the intermediate goods that affect productivity in the goods production function. We now compare our setting with that in the conventional R&D model. To be able to directly compare both models, we simplify the Romer model to that without product variety, and with endogenous productivity similar to human capital a la (Lucas, 1988), and with our AI accumulation.

The representative household is endowed with one unit of time. At an instant in time, the household devotes the fraction of time u to improving its skills, the fraction of time v to developing AI, the fraction of time d to inventing innovation (conventional R&D) which affects the firm's productivity, and the remaining fraction of time $1-u-v-d$ to goods production. We denote m as conventional R&D and the accumulation function of such innovation is as follows:

$$\dot{m} = G(dh)^{\xi} m^{1-\xi}, \tag{15}$$

where $\xi \in (0,1)$, $G > 0$ measures the efficiency of the process of conventional R&D accumulation, and the initial level of $m(0)$ is taken as given.

The representative firm's production function becomes as follows:

$$y = m^{\varepsilon}[(1-s)k]^{1-\alpha-\varepsilon}\{(1-a)[(1-u-v-d)h]^{\beta} + aA^{\beta}\}^{\frac{\alpha}{\beta}}, \tag{16}$$

where $\varepsilon, \alpha, 1-\alpha-\varepsilon \in (0,1)$ and m^{ε} is similar to the firm's productivity F in (7). Other settings are the same as those in the benchmark model. By using the same steps as in the benchmark model, we can derive the equilibrium conditions and the BGP. To more clearly discuss the effect of AI along the transitional dynamics and balanced growth paths, and to compare it with the benchmark model, we undertake a numerical analysis.

To quantify the results, we calibrate the model along the BGP to reproduce the key features of the US economy at annual frequencies. Similarly, we use data for the period 2000–2017. By using the same database as those in the benchmark model (Section 3.1), we can derive that $\gamma = 2.0234\%$, $q = 0.8906$, and $z = 0.1839$. In addition, according to Cooley (1995) we set $\alpha = 1 - 0.4 = 0.6$. Moreover, by setting $\sigma = 2$ and $\rho = 4\%$, we can derive that $u = 0.2515$ and $B = 0.0805$. Similarly, we set $\beta = 1 - (1/0.406)$ and assume that $\phi = \theta = 1/3$ and $a = 0.5$ as those in the benchmark. Here, we further assume that $\varepsilon = 0.09$ and $\xi = 0.5$. We can calibrate $s = 0.1264$, $v = 0.2004$, $d = 0.0578$, $x = 0.1763$, $D = 0.0225$, $G = 0.0019$, and $p = 0.0005$. Finally, by normalizing the initial human capital level at $h(0) = 1$, the household's welfare can be derived from (13i) as $U = -65.2577$. We summarize the related macroeconomic variables in Table 5 (case A).

[10] Related figures are available upon request.

[11] All comparative statics results and the discussions are in the online appendix.

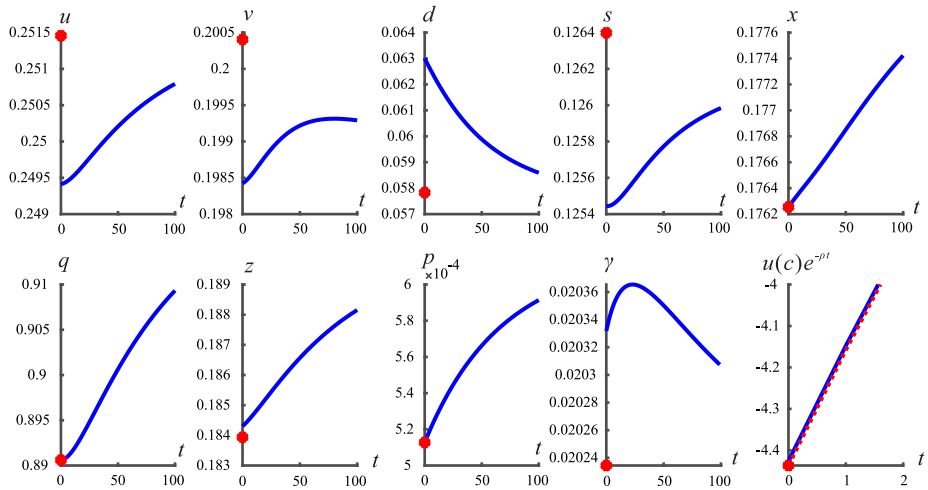

Fig. 4. Transitional dynamics under an increase in productivity in the R&D sector in the model with conventional R&D. Note: The initial point at time 0 is the benchmark case, i.e., case (A) in Table 5. The dotted line in the rightmost graph in the second row is the discounted utility in the benchmark case. To clearly understand the impact, here, we only show the changes for the first 2 periods in the graph of the discounted utility.

We first discuss the impact of changing the parameters related to AI along the transitional dynamics path. To compare with the benchmark model, here we still investigate the transition impact of an increase in the self-accumulation ability of AI and the CES share of AI in the goods production. Furthermore, an increase in efficiency of the process of conventional R&D accumulation (G) will increase the stock of conventional R&D and will increase the productivity of goods production. We thus investigate the transition impact of an increase in G.

Fig. 4 shows that the transition results from a 10% increase in productivity in the R&D sector of which G ranges from 0.0019 to 0.0021, i.e., the transitional dynamics of case (B) in Table 5 with case (A) as the initial point. A higher efficiency of conventional R&D accumulation increases the household's incentive to allocate more time to the R&D sector and thus, d increases immediately. Therefore, the time allocated to the goods, education, and AI sectors decreases the moment G increases. Because of the smaller amount of time devoted to the goods sector and the greater development of R&D, which is beneficial for goods production, the household allocates more physical capital to the goods sector, and thus $1 - s$ increases.

As there is more physical capital and R&D stock in the goods sector, goods production increases over time and thus, the economic growth rate increases in the short run. This also helps the accumulation of AI and physical capital, and increases the household's consumption. That is, the ratios of AI, physical capital, conventional R&D accumulation, and consumption to human capital increase over time, as does household utility. Similar to Fig. 1, an increase in productivity of goods production helps the accumulation of AI, and this contributes to economic growth and public welfare along the transitional dynamics path.

Fig. 5 shows that the transition results from a 10% increase in AI self-accumulation abilities in which D ranges from 0.0225 to 0.0248, i.e., the transitional dynamics of case (C) in Table 5 with case (A) as the initial point. A higher efficiency of AI accumulation increases the household's incentive to allocate more time and resources to the AI sector, and thus v and s increase immediately. Different from Fig. 2, where u decreases immediately, here, u increases immediately, but then decreases below the benchmark level. We find that the time allocated to the R&D and goods sectors decreases immediately and thus, the ratio of conventional R&D accumulation to human capital decreases over time.

Although less time is devoted to working, goods production and economic growth increase in the short run because of greater accumulation of AI and human capital. The increased level of goods production helps the accumulation of AI and physical capital and increases the household's consumption. That is, the ratios of AI, physical capital, and consumption to human capital increase over time, as does household utility. Similar to Fig. 2, a higher AI self-accumulation ability increases the accumulation of AI, economic growth, and public welfare along the transitional dynamics path.

Fig. 6 shows the transition results from a 10% increase in the CES share of AI in goods production in which a ranges from 0.5 to 0.55, i.e., the transitional dynamics of case (D) in Table 5 with case (A) as the initial point. As the production of final goods needs more AI, the household allocates more time and resources to develop AI and thus, v and s increase immediately. Different from Fig. 3, where u decreases immediately, here, u increases immediately but then decreases below the level in the benchmark model. Therefore, we find that the ratio of AI stock to human capital increases over time. Furthermore, the time allocated to the R&D and goods sectors decreases immediately, while d then increases to above the level in the benchmark model. That is, the ratio of the conventional R&D accumulation to human capital first decreases and then increases over time.

The negative effects of less working time and conventional R&D accumulation dominate the positive effects of more AI and human capital accumulation and thus, the economic growth rate declines immediately and then increases to above the level in the benchmark model in the short run. The above negative effects on goods production also influence the accumulation of physical capital and household consumption and thus, q and z are lower than their levels in the benchmark model, as is also the case for

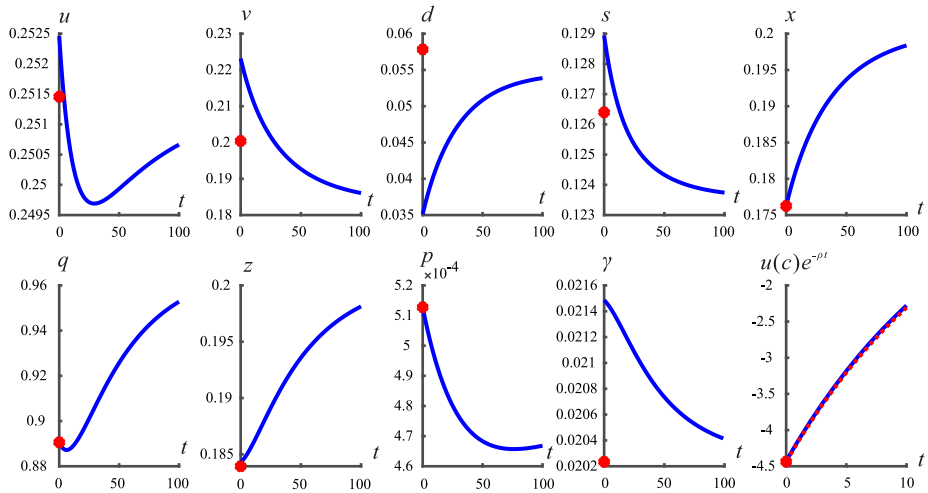

Fig. 5. Transitional dynamics under an increase in AI self-accumulation abilities in the model with conventional R&D. Note: Please refer to Fig. 4. To clearly understand the impact, here, we only show the changes for the first 10 periods in the graph of the discounted utility.

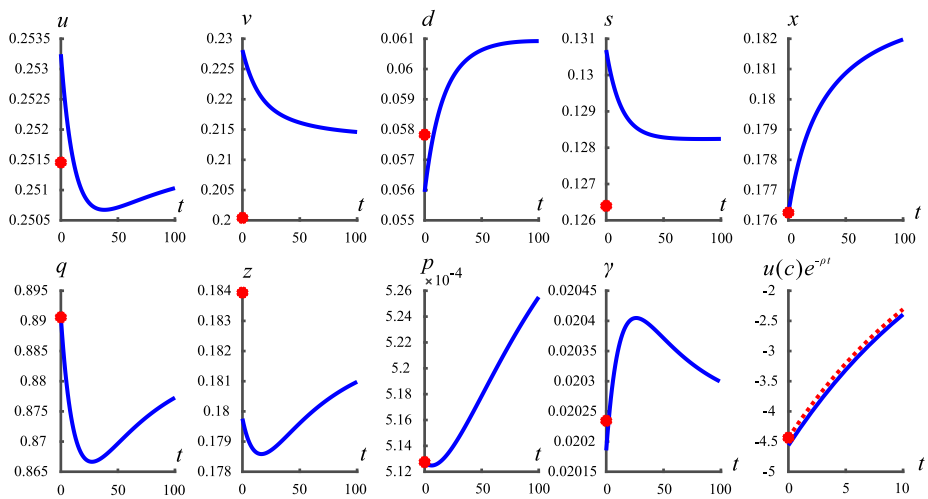

Fig. 6. Transitional dynamics under an increase in the CES share of AI in goods production in the model with conventional R&D. Note: Please refer to Fig. 5.

household utility. In this case, the development of AI may be helpful for short-term economic growth, but it may negatively affect very short-term economic growth. Moreover, people may feel less happy along the transitional dynamics path.

In general, along the transitional dynamics path, the development of AI does help increase the short-term economic growth rate and the short-term welfare of the people if the incentive for the household to accumulate AI comes from the improvement of production efficiency, whether it is from the goods sector or the AI accumulation sector. However, it may negatively affect very short-term economic growth if the incentive for the household to accumulate AI comes from the firms changing the production methods and replacing human labor with AI. In addition, it may further reduce short-term household utility.

Regarding the long-run effects of different parameter values on those variables related to AI, economic growth, and welfare. Similar to Table 2, the household's time allocated to the accumulation of human capital (u) and the economic growth rate along the BGP are unchanged under all cases except for changing B. Except for the new endogenous variables d and p and the new parameters ε and ξ in this model,[12] all comparative static results in the model with conventional R&D are the same as those in the benchmark model, as is the intuition. This implies that introducing conventional R&D accumulation does not influence our main results.

[12] All comparative statics results and the related discussions are in the online appendix.

Table 6
Calibration results of the model with different kinds of physical capital.

u	v	s	e	x	q	z	γ	U
(A) *The benchmark case*								
0.2515	0.2644	0.2678	0.5773	0.1428	0.8906	0.1839	0.0202	−65.2577
(B) F *is increasing from* 0.8175 *to* 0.8992								
0.2515	0.2525	0.2657	0.5766	0.1535	1.0877	0.2250	0.0202	−48.7734
(C) D *is increasing from* 0.0167 *to* 0.0183								
0.2515	0.2398	0.2634	0.5758	0.1659	1.0157	0.2106	0.0202	−53.8405
(D) a *is increasing from* 0.5 *to* 0.55								
0.2515	0.2828	0.2706	0.5782	0.1466	0.8671	0.1786	0.0202	−67.9312

6. Extension: Different kinds of physical capital

We now extend our model to incorporate different kinds of physical capital and check if our main results still hold. Assume that physical capital is decomposed into hardware and software with proportions $1-e$ and e, respectively. Only software can be used to develop AI. Similarly, a fraction s of software is allocated to develop AI and the remaining fraction $1-s$ is allocated to produce goods. That is, the process of AI accumulation, the representative household's flow budget constraint, and the representative firm's production function become as follows, respectively:

$$\dot{A} = D(vh)^{\phi}(esk)^{\theta}A^{1-\phi-\theta}, \tag{17}$$

$$\dot{k} = w(1-u-v)h + r_h(1-e)k + r_s e(1-s)k + r_A A + \pi - c, \tag{18}$$

$$y = F[(1-e)k]^{1-\alpha-\zeta}[e(1-s)k]^{\zeta}\{(1-a)[(1-u-v)h]^{\beta} + aA^{\beta}\}^{\frac{\alpha}{\beta}}, \tag{19}$$

where r_h and r_s are the rental rates for hardware and software, respectively, and $\zeta, \alpha, 1-\alpha-\zeta \in (0,1)$. Other settings are the same as those in the benchmark model.

By using the same steps as in the benchmark model, we can derive the equilibrium conditions and the BGP. To more clearly discuss the effect of AI along the transitional dynamics and balanced growth paths, and to compare it with the benchmark model, we undertake a numerical analysis.

To quantify the results, we calibrate the model along the BGP to reproduce the key features of the US economy at annual frequencies. Similarly, we use data for the period 2000–2017. By using the same database as that in the benchmark model (Section 3.1), we can derive that $\gamma = 2.0234\%$, $q = 0.8906$, and $z = 0.1839$. Moreover, by setting $\alpha = 1-0.3 = 0.7$, $\sigma = 2$, and $\rho = 4\%$, we can derive that $u = 0.2515$ and $B = 0.0805$. Similarly, we set $\beta = 1-(1/0.406)$ and assume that $\phi = \theta = 1/3$ and $a = 0.5$ as in the benchmark model. Here, we further assume that $\zeta = 0.15$. We can calibrate $s = 0.2678$, $v = 0.2644$, $e = 0.5773$, $x = 0.1428$, $D = 0.0167$, and $F = 0.8175$. Finally, by normalizing the initial human capital level at $h(0) = 1$, the household's welfare can be derived from (13i) as $U = -65.2577$. We summarize the related macroeconomic variables in Table 6 (case A).

We first discuss the impact of changing the parameters related to AI along the transitional dynamics path. To compare with the benchmark model, here we still investigate the transition impact of an increase in the parameter related to productivity and the CES share of AI in the production of goods, and the self-accumulation ability of AI. We increase one parameter by 10% at a time. The long-run simulation results are listed in Table 6 (cases B–D, respectively).

The transitional dynamics paths for variables also in the benchmark model under 10% increases in F, D, and a are the same as those in Figs. 1–3, respectively, and the transitional dynamics path for e under each case is the same as that of s. That is, introducing different kinds of physical capital does not change our main results: the development of AI can increase economic growth along the transitional dynamics path, and can benefit household short-run utility if the accumulation of AI is from the increasing productivity in the goods or AI sector, but can be detrimental to household short-run utility if the accumulation of AI is because firms use more AI to replace human labor.

As for the long-run effects of different parameter values on those variables related to AI, economic growth, and welfare. Except for the new endogenous variable e and new parameter ζ in this model,[13] all comparative static results in the model with different kinds of physical capital are the same as those in the benchmark model (Table 2), as is the intuition. This implies that introducing different kinds of physical capital does not influence our main results.

7. Concluding remarks

This paper builds a three-sector endogenous growth model incorporating the self-accumulation ability and the nonrival characteristic of AI, and investigates the effects of the development of AI on economic growth and welfare. However, we did

[13] All comparative statics results and the related discussions are in the online appendix.

not analyze any heterogeneity among skilled workers or jobs, and did not consider any problems associated with AI, such as unemployment or intergenerational transfers. Note that Sachs and Kotlikoff (2012) built an overlapping generation model with smart machines, and emphasized the distributional conflict across generations introduced by a form of capital owned by the elderly that can self-replicate and substitute for the labor of the young. It would be interesting to study the implications for income distribution across generations in this framework. We defer such analyses to future research.

CRediT authorship contribution statement

Chia-Hui Lu: Conceptualization, Methodology, Software, Writing - original draft, Writing - review & editing.

Appendix A. Calculation details

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References

Acemoglu, D., Restrepo, P., 2017. Low-skill and high-skill automation. NBER Working Paper 24119.
Acemoglu, D., Restrepo, P., 2018a. Artificial Intelligence, Automation and Work. NBER Working Paper 24196.
Acemoglu, D., Restrepo, P., 2018b. The race between man and machine: implications of technology for growth, factor shares and employment. Amer. Econ. Rev. 108, 1488–1542.
Aghion, P., Howitt, P., 1992. A model of growth through creative destruction. Econometrica 60, 323–351.
Aghion, P., Jones, B., Jones, C., 2017. Artificial Intelligence and Economic Growth. NBER Working Paper 23928.
Autor, D., 2015. Why are there still so many jobs? The history and future of workplace automation. J. Econ. Perspect. 29, 3–30.
Berg, A., Buffie, E., Zanna, L.-F., 2018. Should we fear the robot revolution? (the correct answer is yes). J. Monetary Econ. 97, 117–148.
Bond, E., Wang, P., Yip, C., 1996. A general two sector model of endogenous growth with human and physical capital: Balanced growth and transitional dynamics. J. Econom. Theory 68, 149–173.
Chirinko, R., Mallick, D., 2017. The substitution elasticity, factor shares, and the low-frequency panel model. Amer. Econ. J.: Macroecon. 9, 225–253.
Cooley, T., 1995. Frontiers of Business Cycle Research. Princeton University Press, Princeton.
Fernald, J., Jones, C., 2014. The future of US economic growth. Amer. Econ. Rev. 104, 44–49.
Hémous, D., Olsen, M., 2016. The Rise of the Machines: Automation, Horizontal Innovation and Income Inequality. Mimeo.
Kydland, F., Prescott, E., 1991. Hours and employment variation in business cycle theory. Econom. Theory 1, 63–81.
Lucas, R., 1988. On the mechanics of economic development. J. Monetary Econ. 22, 3–42.
Lucas, R., 1990. Supply-side economics: An analytical review. Oxf. Econ. Pap. 42, 293–316.
Mino, K., 2001. Indeterminacy and endogenous growth with social constant returns. J. Econom. Theory 97, 203–222.
Morikawa, M., 2017. Firms' expectations about the impact of AI and robotics: Evidence from a survey. Econ. Inq. 55, 1054–1063.
Pratt, G., 2015. Is a cambrian explosion coming for robotics?. J. Econ. Perspect. 29, 51–60.
Romer, P., 1990. Endogenous technological change. J. Polit. Econ. 98, S71–S102.
Sachs, J., Kotlikoff, L., 2012. Smart Machines and Long-Term Misery. NBER Working Paper 18629.
Silver, D., et al., 2017. Mastering the game of go without human knowledge. Nature 550, 354–359.